Contribution

According to the group assignment policy, each team member needed to complete the lab tasks independently. I completed, checked, and documented the full workflow for the Module 02 lab. My main contributions included:

1. Environment Configuration and Package Verification

I set up the workspace and checked the required packages, including pandas (v2.2.3), NumPy (v2.1.3), matplotlib.pyplot, and sklearn.datasets.

2. Data Loading and DataFrame Operations

I loaded the Iris dataset using load_iris() and created a pandas DataFrame with 150 instances, four feature columns, and target species labels. I also used .info() and .head() to check the data.

3. Statistical Calculation and Verification

I calculated statistics for each species and calculated the mean (5.84 cm) and standard deviation (0.83 cm) of sepal length. I also used assert statements to check the results.

4. Data Visualization

I created a matplotlib bar chart to show the number of samples for *setosa*, *versicolor*, and *virginica*.

5. Deliverable Compilation and Repository Management

I exported the completed notebook results as L02_YilinLeng_ITAI1371.pdf, wrote my reflection and contribution logs, created the deliverables/yilinleng-pdfs feature branch in the shared ITAI1371-L02-ML-Dev-Tools repository, and submitted a Pull Request to the main branch.